Derivation of Modal Model

The state PDE with viscosity added, and spatial actuator profile $\psi(x,y)$, with speed $c^2>0$, damping coefficient $\gamma>0$, and control parameter $b(t)$, is given by

$$\begin{cases} u_{tt}(x,y,t) = c^2\Delta u(x,y,t) - \gamma u_t(x,y,t) + b(t)\psi(x,y) & \text{for } t\geq 0,\ (x,y)\in[0,1]\times[0,1] =: \Omega \\ \text{with } \quad u(x,y,t) = 0 \text{ on } \partial\Omega \\ \text{and } u(x,y,0) = u_0 \text{ on } \Omega \text{ at } t=0. \end{cases}$$

The Dirichlet Laplacian Eigenfunctions $\phi_{mn}(x,y)$ and corresponding eigenvalues λ_{mn} are given by

$$\phi_{mn}(x,y) = 2\sin(m\pi x)\sin(n\pi y)$$
$$\lambda_{mn} = \pi^2(m^2+n^2)$$

So we can express $u(x,y,t)$ as a time-dependent linear combination of the spatial modes $\phi_{mn}(x,y)$, with time-dependent unknown coefficients $q_{mn}(t)$:

$$u(x,y,t) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} q_{mn}(t)\,\phi_{mn}(x,y)$$

Now, we substitute this back into the original PDE:

$$\frac{\partial^2}{\partial t^2}\left(\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} q_{mn}(t)\,\phi_{mn}(x,y)\right) = c^2\Delta\left(\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} q_{mn}(t)\,\phi_{mn}(x,y)\right) - \gamma\frac{\partial}{\partial t}\left(\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} q_{mn}(t)\,\phi_{mn}(x,y)\right) + b(t)\psi(x,y)$$

$$\Rightarrow \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \ddot{q}_{mn}(t)\,\phi_{mn}(x,y) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} c^2 q_{mn}(t)\left(\Delta\phi_{mn}(x,y)\right) - \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \gamma\dot{q}_{mn}(t)\phi_{mn}(x,y) + b(t)\psi(x,y)$$

$$\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \ddot{q}_{mn}(t)\,\phi_{mn}(x,y) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} c^2 q_{mn}(t)\left(-\lambda_{mn}\phi_{mn}(x,y)\right) - \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \gamma\dot{q}_{mn}(t)\phi_{mn}(x,y) + b(t)\psi(x,y)$$

Finally, we project onto the mode $\phi_{mn}(x,y)$, using the orthogonality condition

$$\langle \phi_{mn}, \phi_{jk}\rangle = \delta_{mn,jk}$$

$$\Rightarrow \ddot{q}_{mn}(t) = -c^2\lambda_{mn}q_{mn}(t) - \gamma\dot{q}_{mn}(t) + b(t)\left(\int_0^1\int_0^1 \psi(x,y)\,\phi_{mn}(x,y)\,dx\,dy\right)$$

$$\Rightarrow \boxed{\begin{aligned} &\ddot{q}_{mn}(t) + \gamma\dot{q}_{mn}(t) + \omega_{mn}q_{mn}(t) = \beta_{mn}b(t) \\ &\text{where} \\ &\omega_{mn} := c^2\lambda_{mn}, \text{ and} \qquad (m,n\in\mathbb{N}_0) \\ &\beta_{mn} := \left(\int_0^1\int_0^1 \psi(x,y)\,\phi_{mn}(x,y)\,dx\,dy\right) \end{aligned}}$$

Derivation of LQR matrices

Truncation & Rewriting as 1st-Order System

Truncating at M modes in each coordinate direction, and "flattening" the double indexes $(1,1), \dots, (M,M)$ to $1, \dots, N = M^2$, we define the state vectors $\vec{q}(t)$ and $\vec{p}(t)$ as

$$\vec{q}(t) = \begin{bmatrix} q_1(t) \\ \vdots \\ q_N(t) \end{bmatrix} ; \vec{p}(t) := \dot{\vec{q}}(t) = \begin{bmatrix} \dot{q}_1(t) \\ \vdots \\ \dot{q}_N(t) \end{bmatrix}$$

$\Rightarrow$ For all $n \in \{1, \dots, N\}$,

$$\dot{q}_n(t) = p_n(t)$$

$$\dot{p}_n(t) = \ddot{q}_n(t) = -\omega_n q_n(t) - \gamma \dot{q}_n(t) + \beta_n b(t) = -\omega_n q_n(t) - \gamma p_n(t) + \beta_n b(t)$$

In matrix/vector form:

$$\dot{\vec{q}}(t) = \underline{\underline{0}}_N \vec{q}(t) + \underline{\underline{I}}_N \vec{p}(t) + b(t)\vec{0}$$

$$\dot{\vec{p}}(t) = \underbrace{\begin{bmatrix} -\omega_1 & & & \\ & -\omega_2 & & \\ & & \ddots & \\ & & & -\omega_N \end{bmatrix}}_{:= -(\underline{\underline{\Omega}}_N)} \vec{q}(t) + (-\gamma \underline{\underline{I}}_N)\vec{p}(t) + b(t) \underbrace{\begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}}_{:= \vec{\beta}}$$

where $\underline{\underline{\Omega}}_n := \mathrm{diag}(\omega_1, \omega_2, \dots, \omega_N)$.

Combining it all

Let the state $\vec{x}(t) := \begin{bmatrix} \vec{q}(t) \\ \vec{p}(t) \end{bmatrix}$.

$$\Rightarrow \boxed{\dot{\vec{x}} = \underline{\underline{A}}\vec{x} + \vec{B}b \quad \text{where} \quad A = \begin{bmatrix} 0 & \underline{\underline{I}}_N \\ -\underline{\underline{\Omega}}_N & -\gamma \underline{\underline{I}}_N \end{bmatrix} ; \vec{B} = \begin{bmatrix} \vec{0}_N \\ \vec{\beta} \end{bmatrix}}$$

Cost & Infinite-Horizon LQR

We want to penalize the displacement according to modal potential energy, and the velocity according to kinetic energy.

The energy of the membrane at a given time is

$$E(t) = \frac{1}{2} \iint (u_t^2 + c^2 u_x^2)\, dx$$

Recall that

$$u_t = \sum_n \dot{q}_n(t)\, \phi_n(x,y) \quad \text{(this yields } \underline{\text{kinetic energy}}\text{)}$$

and $$u_x = \sum_n q_n(t)\, (\phi_n)_x(x,y) \quad \text{(this yields } \underline{\text{potential energy}}\text{)}$$

etc.

Thus, it makes physical sense to minimize over the stuff we have control over; i.e., the q_n coefficients.

$\Rightarrow$ <u>Potential Energy Cost</u>: $\alpha\left(\sum \omega_n q_n^2\right) = \alpha(\vec{q}_n^T \Omega_N \vec{q}_n) \quad (\alpha > 0)$

<u>Kinetic Energy</u>: $\beta_v\left(\sum \dot{q}_n^2\right) = \beta_v\left(\sum p_n^2\right) = \beta_v(\vec{p}^T I_N \vec{p}) \quad (\beta_v > 0)$

We also want to make it expensive to move the actuator, so we let R be some cost to actually move the mechanical part away from equilibrium ($R \approx 0.05$ is great).

$$J(b) = \int_0^\infty \left[\alpha(\vec{q}_n^T \Omega_N \vec{q}_n) + \beta_v(\vec{p}^T I_N \vec{p}) + R\, b(t)^2\right] dt$$

$$= \int_0^\infty [\vec{q}(t)^T \ \vec{p}(t)^T] \begin{bmatrix} \alpha \underline{\underline{\Omega}}_N & \underline{\underline{0}}_N \\ \underline{\underline{0}}_N & \beta_v \underline{\underline{I}}_N \end{bmatrix} \begin{bmatrix} \vec{q}(t) \\ \vec{p}(t) \end{bmatrix} + R\, b(t)^2 \, dt$$

$$= \int_0^\infty \vec{x}(t)^T \underline{\underline{Q}}\, \vec{x}(t) + [b(t)]^T \underline{\underline{R}}\, [b(t)] \, dt$$

where

$$\underline{\underline{Q}} = \begin{bmatrix} \alpha \underline{\underline{\Omega}}_N & \underline{\underline{0}}_N \\ \underline{\underline{0}}_N & \beta_v \underline{\underline{I}}_N \end{bmatrix} ; \ \underline{\underline{R}} = [R].$$

LQR Feedback

The LQR feedback (in infinite horizon) has the form

$$b(t) = -\underline{\underline{K}}\,\vec{x}(t),$$

where

$$\underline{\underline{K}} = \underline{\underline{R}}^{-1}\,\underline{\underline{B}}^{T}\,\underline{\underline{P}}$$

and $\underline{\underline{P}}$ solves the continuous algebraic Ricatti equation

$$\underline{\underline{A}}^{T}\underline{\underline{P}} + \underline{\underline{P}}\,\underline{\underline{A}} - \underline{\underline{P}}\,\underline{\underline{B}}\,\underline{\underline{R}}^{-1}\underline{\underline{B}}^{T}\underline{\underline{P}} + \underline{\underline{Q}} = \underline{\underline{0}}$$